Lecture 37 — Why did the system behave that way?

Fri Nov 20

Last updated: August 18, 2026.

The class outline for this day will be posted before class.

Reading: Strang §6.1, Lab 4, and Lachniet Ch. 5. Related objectives: Lectures.

Learning objectives

By the end of class, students should be able to:

1. Compute eigenvalues and eigenvectors of a general matrix with software.
2. Verify $Av \approx \lambda v$ numerically.
3. Use dominant eigenbehavior to explain a simulated system.
4. Interpret complex eigenvalues as rotation/oscillation behavior.
5. Explore a symmetric-matrix or normal-mode example and observe orthogonal eigenvectors.